

(Ir)responsible Takeovers

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Introduction

Investors frequently profess to care about social externalities

Climate, pollution, worker welfare, geopolitics, free speech ...

Key question: Do (would) these preferences affect decisions?

Change in corporate control = clearest point of shareholder influence

This paper: Take investors at their word on caring about externalities

Market for corporate control w/ externality-minded shareholders

In many ways, paper is a thought experiment

what would world like this look like?

are equilibrium outcomes desirable?

Economics: Free-riding in social responsibility, free-riding in tendering

Overview

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(1) Sharp and clean characterization of equilibrium outcomes

Balance of social preferences btw target shareholders & bidder matters

Balanced preferences $\rightarrow$ desirable outcomes

Overview

Economics: Free-riding in social responsibility, free-riding in tendering

(1) Sharp and clean characterization of equilibrium outcomes

Balance of social preferences btw target shareholders & bidder matters

Balanced preferences → desirable outcomes

(2) Use sharp characterization to explore:

- a. Bidder choice of externalities
 - b. Trade with purely financial agents
 - c. Equity vs cash offers
 - d. Leveraged offers (LBOs)
 - e. Legal protections for minority shareholders
- (paper has further results)

Model: Externalities + Bagnoli-Lipman 1988 (Grossman-Hart 1980)

Target: N shareholders, per-share pecuniary value v_0 and externality ϕ_0

K of N shares needed for control

Bidder: $v_0 \rightarrow v_1$ and $\phi_0 \rightarrow \phi_1$, conditional offer p per share

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Preferences

$q \equiv$ prob takeover succeeds if an individual shareholder holds out

$q + \Delta \equiv$ prob takeover succeeds if an individual shareholder tenders

Shareholder retains $\rightarrow v_0 + \alpha\phi_0 + q(v_1 + \alpha\phi_1 - v_0 - \alpha\phi_0)$

Divested shareholder weights externality by $\eta \leq \alpha$

Shareholder tenders $\rightarrow v_0 + \alpha\phi_0 + (q + \Delta)(p + \eta\phi_1 - v_0 - \alpha\phi_0)$

Bidder weights acquired externalities by δ

Bidder's payoff = $\Pr(\text{success}) E[\text{shares acquired} | \text{success}] (v_1 + \delta\phi_1 - p)$

Motivating examples

Takeover socially (in)efficient $\Leftrightarrow v_1 + \phi_1 (<) > v_0 + \phi_0$

BP has pursued relatively aggressive decarbonization policies

Likely possible to increase cash flows ($v_1 > v_0$)
by reversing decarbonization efforts ($\phi_1 \ll \phi_0$)

?Socially inefficient takeover succeeds because of freeriding?

Exxon hasn't invested much in decarbonization policies

Likely possible to improve externalities ($\phi_1 \gg \phi_0$)
though at cost to cash flows ($v_1 < v_0$)

?Socially efficient takeover fails because of freeriding?

Assume α large enough that:

$$\text{sign}(v_1 + \alpha\phi_1 - v_0 - \alpha\phi_0) = \text{sign}(v_1 + \phi_1 - v_0 - \phi_0)$$

Freeriding

Two freeriding problems

Grossman-Hart: each shareholder wants to freeride on others' tendering

Externalities: each shareholder has little influence over externalities

Problems are different

GH: Tendering $\rightarrow$ give up on v_1

Externalities: Tendering $\rightarrow$ still exposed to ϕ_1

Bidder profits from v_1 but not necessarily from ϕ_1

Formally: Can't just absorb ϕ 's into v 's

Shareholder tenders $\rightarrow v_0 + \alpha\phi_0 + (q + \Delta)(p + \eta\phi_1 - v_0 - \alpha\phi_0)$

Shareholder retains $\rightarrow v_0 + \alpha\phi_0 + q(v_1 + \alpha\phi_1 - v_0 - \alpha\phi_0)$

Ownership-invariant shareholder preferences

Leading special case:

shareholders have ownership-invariant preferences ($\eta = \alpha$)

bidder doesn't care about externalities ($\delta = 0$)

Remark: Unified shareholders $\rightarrow$ first-best

Question: What happens w/ dispersed shareholders?

Ownership-invariant shareholder preferences

Leading special case:

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Remark: Unified shareholders $\rightarrow$ first-best

Question: What happens w/ dispersed shareholders?

Result:

Market for corporate control works well

Externalities fully internalized

Socially inefficient takeovers fail

Socially efficient takeovers succeed w/ same prob as if externalities were pecuniary cash flows ($\tilde{v}_\omega = v_\omega + \alpha\phi_\omega$)

Socially inefficient takeovers fail

Shareholder tenders $\rightarrow v_0 + \alpha\phi_0 + (q + \Delta)(p + \alpha\phi_1 - v_0 - \alpha\phi_0)$

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$$\begin{aligned}\text{Willingness-to-Tender } (WTT) &= \Delta(v_1 + \alpha\phi_1 - v_0 - \alpha\phi_0) - (q + \Delta)(v_1 - p) \\ &= \Delta s - (q + \Delta)(v_1 - p)\end{aligned}$$

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Case: $\phi_1 = \phi_0 = 0$, Grossman-Hart/Bagnoli-Lipman

Individual: “public good” contribution $\Delta(v_1 - v_0)$ vs cost $(q + \Delta)(v_1 - p)$

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Socially inefficient takeovers fail

$s < 0$ and bidder bids $p \leq v_1 \Rightarrow \text{WTT} < 0$

Socially inefficient takeovers fail: Illustration

Reversing BP's decarbonization: $v_1 > v_0$ and $\phi_1 \ll \phi_0$

Might conjecture: Takeover with $p > v_0$ succeeds

b/c SHs compare benefit $p - v_0 > 0$ w/ small influence on $\alpha\phi_1 - \alpha\phi_0 \ll 0$

Instead, GH freeriding blocks takeover

SHs compare $p - v_1 < 0$ w/ small influence on $v_1 + \alpha\phi_1 - v_1 - \alpha\phi_0 < 0$

Socially efficient takeovers

$\gamma \equiv$ equilibrium tendering probability

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$$q(\gamma) = \sum_{j=K}^{N-1} \binom{N-1}{j} \gamma^j (1-\gamma)^{N-1-j}$$

$$\Delta(\gamma) = \binom{N-1}{K-1} \gamma^{K-1} (1-\gamma)^{N-K}$$

Socially efficient takeovers

$\gamma \equiv$ equilibrium tendering probability

(mixing) $WTT = \Delta s - (q + \Delta)(v_1 - p) = 0$

Note:

GH freeriding $v_1 - p =$ bidder profits/share $v_1 - p$

Socially efficient takeovers

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Offer $p \rightarrow$ tender probability γ

(Offer $p \nearrow$ from $v_0 + \alpha\phi_0 - \alpha\phi_1$ to $v_1 \Rightarrow$ tendering prob $\gamma \nearrow$ from 0 to 1)

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Bidder picks p and hence γ to maximize

$$\Pr(\text{success}) E[\text{shares acquired} | \text{success}] (v_1 - p)$$

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$$\Pr(\text{success}) E[\text{shares acquired} | \text{success}] \frac{\Delta}{q + \Delta} s$$

Economics: $\uparrow$ social value $\rightarrow$ bidder can $\downarrow$ offer

Socially efficient takeovers

$\gamma \equiv$ equilibrium tendering probability

(mixing) $WTT = \Delta s - (q + \Delta)(v_1 - p) = 0$

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Equilibrium γ depends only on sign of s

Equilibrium $\gamma = \frac{K}{N}$, takeover success prob $\approx \frac{1}{2}$

Socially inefficient takeovers sometimes succeed: Illustration

Greening Exxon: $v_1 < v_0$ and $\phi_1 \gg \phi_0$

Might conjecture: Takeover impossible

b/c SHs compare $p - v_0 < v_1 - v_0 < 0$ w/ small influence on $\alpha\phi_1 - \alpha\phi_0 > 0$

Instead:

SHs compare $p - v_1 \approx -\epsilon$ w/ small influence on $v_1 + \alpha\phi_1 - v_1 - \alpha\phi_0 > 0$

In particular: provided p high enough that $p + \alpha\phi_1 > v_0 + \alpha\phi_0$ then

tendering $v_0 + \alpha\phi_0 + (q + \Delta)(p + \alpha\phi_1 - v_0 - \alpha\phi_0)$ potentially beats

retention $v_0 + \alpha\phi_0 + q(v_1 + \alpha\phi_1 - v_0 - \alpha\phi_0)$

The balance of shareholder and bidder preferences

Result: SH-preferences ownership-invariant + bidder doesn't care
Externalities fully internalized

Economics: shareholder and bidder preferences balance

In general

$$WTT = \Delta s - (q + \Delta) (v_1 + (\alpha - \eta) \phi_1 - p)$$

If bidder puts weight $\delta = \alpha - \eta$ on acquired externalities

$$\text{GH freeriding } v_1 + (\alpha - \eta) \phi_1 - p = \text{bidder "profits"/share } v_1 + \delta - p$$

$\Rightarrow$ full-internalization result generalizes

Aside: Special case of $\eta = 0$ (pure warm-glow SHs) and $\delta = \eta$ is obvious!

Imbalanced preferences: Too little weight on divested externalities

For simplicity, focus on cases in which bidder doesn't care ($\delta = 0$):

$$v_1 - p = \frac{\Delta}{q + \Delta} (v_1 + \phi_1 - v_0 - \phi_0) - (\alpha - \eta) \phi_1$$

Negative externalities: $\eta < \alpha$ and $\phi_1 < 0$: $\Pr(\text{success}) \rightarrow 1$

regardless of other parameters

Economics: Shareholders strongly motivated to divest

Socially inefficient takeovers succeed if $\phi_1 < 0$

Positive externalities: $\eta < \alpha$ and $\phi_1 > 0$: $\Pr(\text{success}) \rightarrow 0$

regardless of other parameters

Economics: Shareholders strongly motivated to retain

Socially efficient takeovers fail if $\phi_1 > 0$

Opposite problems if bidder + SHs collectively care too much ($\eta + \delta > \alpha$)

Consequentialist bidder

Consider: consequentialist bidder, weight w_B on externalities

For illustration: consequentialist SHs, $\alpha = \eta$

Assume $N\alpha + w_B = 1$ (o/w, inefficiency even under full coordination)

Bidder benefit from social improvement $\Phi_1 - \Phi_0 \perp \#$ acquired shares

→ social improvement $\Phi_1 - \Phi_0$ resembles a private benefit

→ possible for socially costly ($V_1 + \Phi_1 - V_0 - \Phi_0 < 0$) to succeed

Conjecture: if all successful takeovers followed by a freezeout merger at offer price p , success θ ...

...as $\theta \rightarrow 1$, then recover balanced-preferences result

Taking stock

So far: Equilibrium characterization

Theorem 1 + corollaries

Balanced preferences $\rightarrow$ market for control works well

Imbalanced preferences $\rightarrow$ socially inefficient outcomes

Next: Use analysis to explore:

- a. Bidder choice of externalities
- b. Trade with purely financial agents
- c. Equity vs cash offers
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Bidder choice of externalities

So far: v_1 and ϕ_1 exogenous

What if bidder chooses from $\left\{ v_1 \left(\tilde{\phi}_1 \right), \tilde{\phi}_1 \right\}$?

Socially desired choice: $\phi_1^{**} \equiv \arg \max_{\tilde{\phi}_1} v_1 \left(\tilde{\phi}_1 \right) + \alpha \tilde{\phi}_1$

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Prop 2(a): Balanced preferences $\rightarrow$ bidder picks ϕ_1^{**}

Under balanced preferences, externalities internalized

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Prop 2(a): Balanced preferences $\rightarrow$ bidder picks ϕ_1^{**}

Under balanced preferences, externalities internalized

Prop 2(b): $\eta+\delta<\alpha \rightarrow$ bidder picks $\tilde{\phi}_1<\phi_1^{**}$

Bidder facilitates takeover by proposing worse externalities, strengthening divestment motive

Trade with purely financial agents

Consider trade btw social target SH's & purely-fin agents ($\alpha = \eta = 0$)

Focus on $\delta = 0$ case

Prop 3(a): No trade if preferences balanced ($\eta = \alpha$)

Outcomes $\sim$ efficient, trade lowers efficiency, no direct gain to divestment

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Prop 3(b): If preferences $\eta < \alpha$ then trade can improve social efficiency

Eg: $s > 0$, $\phi_1 > 0$, $v_1 > v_0$. Status quo $\rightarrow$ socially desirable takeover fails

Financial ownership $\rightarrow$ takeover succeeds. Trade occurs if, eg, ϕ_0 low

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Financial ownership $\rightarrow$ takeover succeeds. Trade occurs if, eg, ϕ_0 low

Prop 3(c): If preferences $\eta < \alpha$ then trade can harm social efficiency

Eg: $s > 0$ & $\phi_1 < 0$. Status quo $\rightarrow$ socially desirable takeover succeeds

Financial ownership $\rightarrow$ takeover fails if $v_1 < v_0$. Trade occurs if, eg, ϕ_0 low

Cash vs equity offers

So far, cash offers. Inefficiencies from eagerness/reluctance to divest

Consider equity offers, which dampen divestment motives

Focus on $\eta = 0$: avoid target SH's prefs over bidder externalities

Bidder, N_B shares; offers e newly-issued shares to tendering SH's

Per-share financial value and externalities: $\frac{V_B + jv_1}{N_B + ej}$ & $\frac{\Phi_B + j\phi_1}{N_B + ej}$

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Prop 4(a): Balanced preferences $\rightarrow$ bidder indifferent btw cash & equity

Even absent externalities this is new to literature

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Prop 4(a): Balanced preferences $\rightarrow$ bidder indifferent btw cash & equity

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Prop 4(b): $\delta < \alpha \rightarrow$ bidder prefers cash if $\Phi_B, \Phi_1 < 0$

Cash offers benefit from target SH desire to divest; equity surrenders this

Prop 4(b): $\delta < \alpha \rightarrow$ bidder prefers equity if $\Phi_B, \Phi_1 > 0$

Cash offers suffer from target SH desire to retain; equity avoids this

LBOs/leveraged offers

Bidder takes loan commitment backed by post-acquisition target shares

If bidder gains control, uses control to merge target

Assume balance preferences (market works well w/out leverage)

LBOs/leveraged offers

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Assume balance preferences (market works well w/out leverage)

Prop 5(a): $s > 0$ & $\phi_1 > \phi_0$: leverage $\rightarrow$ efficiency $\uparrow$

Generalizes Mueller-Panunzi (2004): LBOs mitigate GH freeriding

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Generalizes Mueller-Panunzi (2004): LBOs mitigate GH freeriding

Prop 5(c): $s < 0$: leverage $\rightarrow$ efficiency $\downarrow$

Consider case $v_1 > v_0$ but $v_1 + \alpha\phi_1 < v_0 + \alpha\phi_0$

Bid $p = v_0 + \epsilon$ with debt $d = v_1 - v_0 \rightarrow$ tender w/ prob 1 is an equilibrium

Tender $\rightarrow p + \alpha\phi_1 = v_0 + \epsilon + \alpha\phi_1$. Retain $\rightarrow v_1 - d + \alpha\phi_1 = v_0 + \alpha\phi_1$

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Tender $\rightarrow p + \alpha\phi_1 = v_0 + \epsilon + \alpha\phi_1$. Retain $\rightarrow v_1 - d + \alpha\phi_1 = v_0 + \alpha\phi_1$

Legal protections for minority shareholder: $v_1 - d > v_0$, i.e., $d < v_1 - v_0 = s$

Absent externalities, stops leverage being mis-used

With externalities: social inefficiency occurs when $v_1 + \alpha\phi_1 - d < v_0 + \alpha\phi_0$

Normatively, the appropriate legal restriction is $d < s$

Legal protections for minority shareholders

Even w/out leverage, legal protections for minority shareholders matter

Recall: balance prefs $\rightarrow \sim$ efficiency

But retaining SH can likely sue to recover v_0 (& surrender share)

Litigation attractive if $v_0 + \eta\phi_1 > v_1 + \alpha\phi_1$

Litigation succeeds with prob $\sigma > 0$ (parameter)

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Prop 6: If litigation cond is satisfied then takeover fails as $N \rightarrow \infty$

Ex post litigation forces bidder to raise offer

Bidder partially does so, but equilibrium tendering prob $\downarrow$

Normatively, the appropriate litigation hurdle is $v_0 + \alpha\phi_0 > v_1 + \alpha\phi_1$

Conclusion

What takeovers occur if shareholders care about externalities?

If shareholder and bidder preferences “balance” then outcomes are good

But if SHs have warm-glow preferences,
not fully offset by bidder pro-social preferences

Takeovers that leave -ve externalities succeed

Takeovers that leave +ve externalities fail

Use equilibrium characterization to explore:

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